
Methodological Landscape in IO

Econ 650

Comparing “Structural” vs. “Reduced Form” using Hastings (gasoline stations) & Nevo (cereal)

- What if I’m not confident about what strategic model the firms are playing?
 - Hastings: results might still be robust
 - Nevo: potentially problematic
- What if I wanted to estimate what would happen if a different merger occurred (e.g. ARCO + Shell)?
 - Hastings: cannot say anything (that quasi-experiment did not occur)
 - Nevo: can simulate mergers before they occur

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■ Structural

- ❑ Model transparent/well-specified, but may/may not match real-world institutions closely
- ❑ Not clear how much (small) mis-specifications matter
- ❑ Allows to do “counterfactual experiments”

■ Reduced-form

- ❑ Do not need “correct” model of demand or price competition
- ❑ Need assumed exogeneity to hold
 - May want to write down a formal model, of say choice of vertical structure, so you know what “hidden” assumptions you’re making
- ❑ Cannot do counterfactual experiments

Empirical Approaches & Their Tools

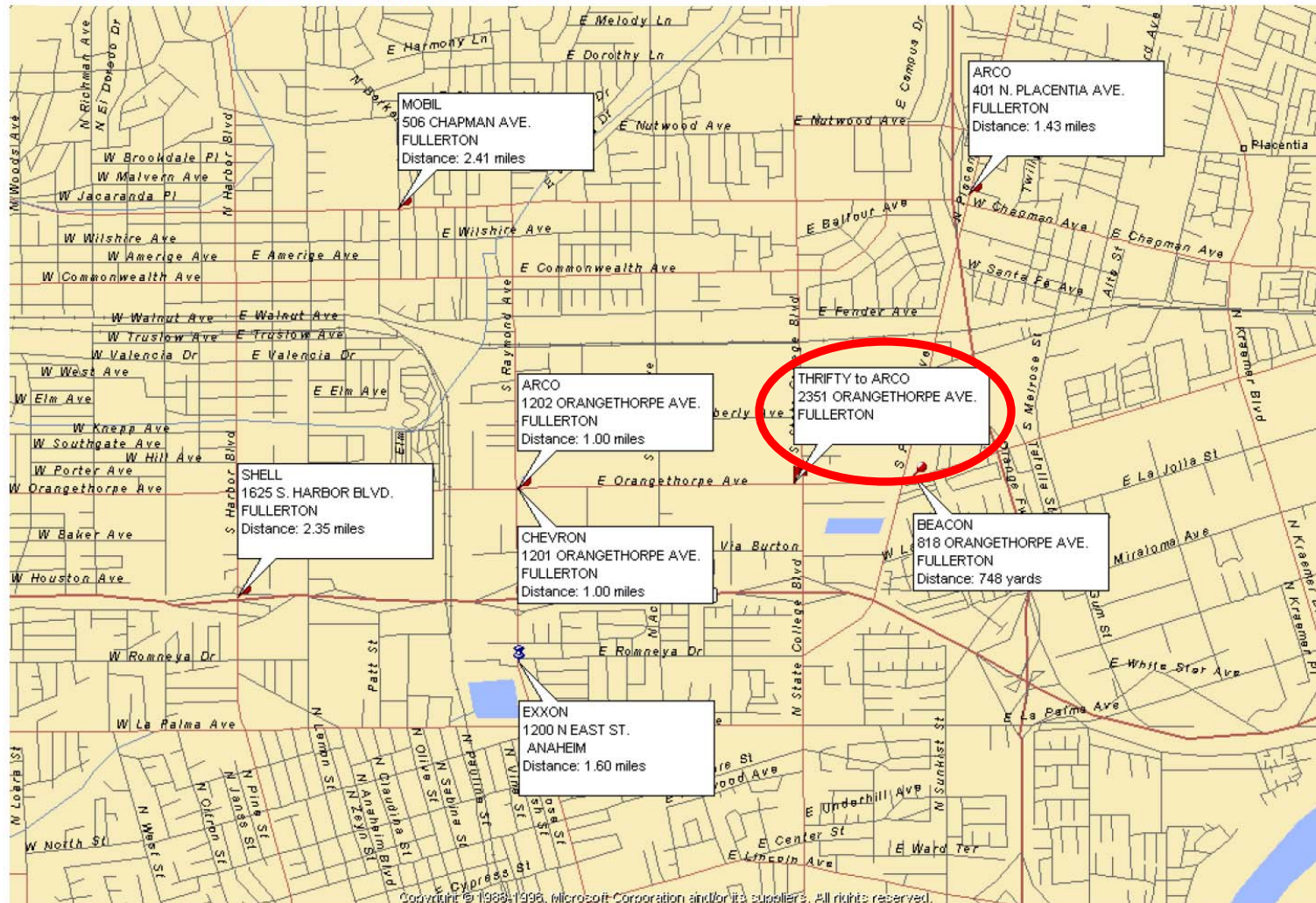
- “Reduced-form approaches”
 - Quasi-experiments, difference-in-difference, regression discontinuity, some instrumental variables
 - “Cute” / “Clever”
 - Shows up in seminar lingo: “what’s your empirical strategy / identification strategy / research design / control group ?”
- Structural Approaches
 - Theory-based models: agents (firms, people) are optimizing
 - Model has underlying parameters to recover
 - Use of an economic theory model to empirically recover parameters (e.g. costs or preferences)
- *I’m going to defend both!*

Nevo & Whinston: Specifically on the case of Merger Analysis

- A&P would have researcher estimate effect of cereal mergers based on “average treatment effect” of past mergers with exogenous variation (e.g. gasoline stations)

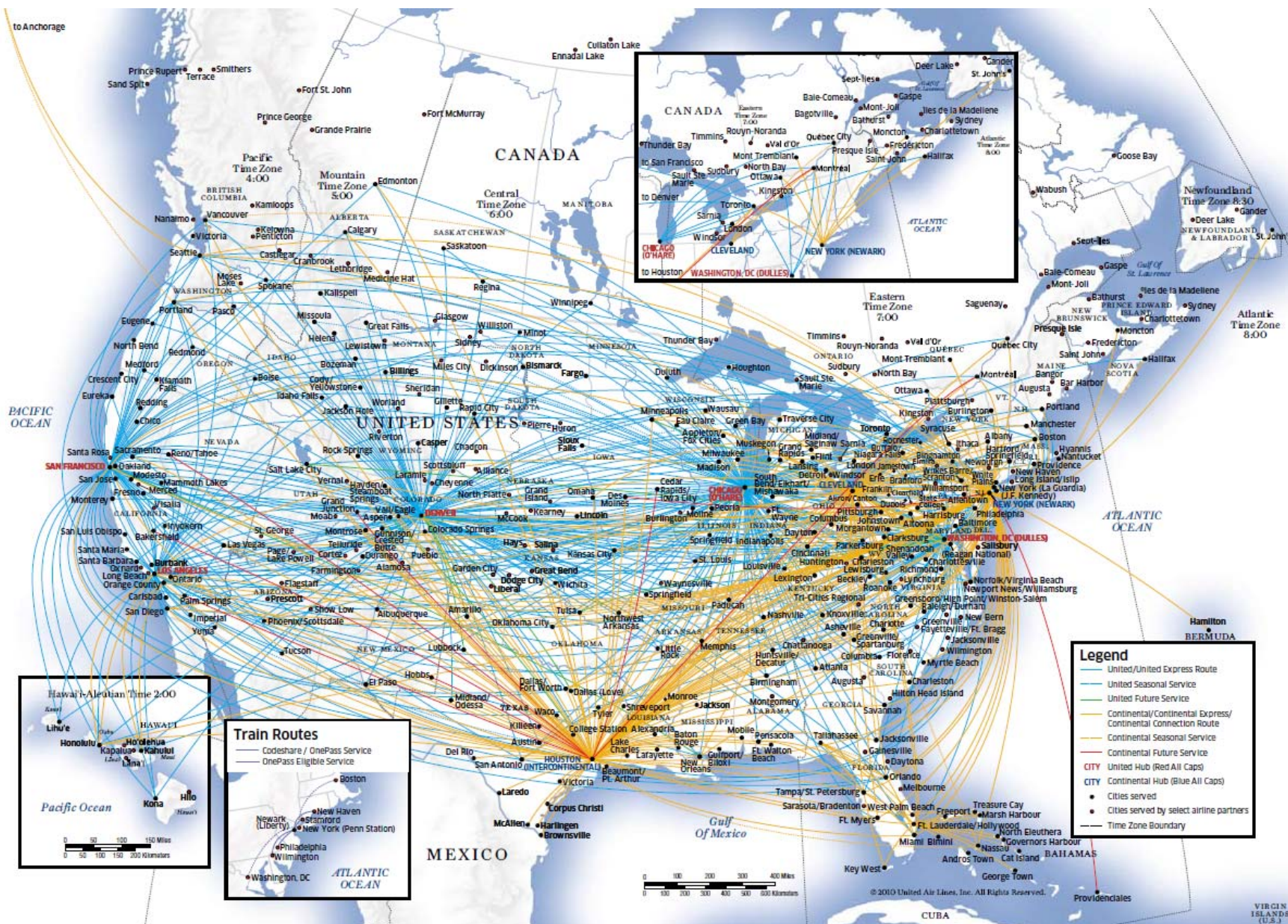
Can We Extrapolate Results from this to...

GRAPH II: Sample Thrifty Station in Fullerton



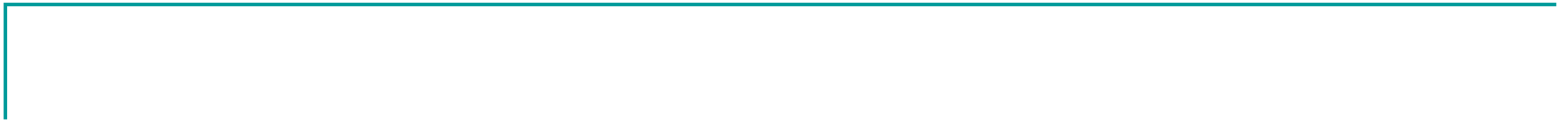
Graph of sample stations in Fullerton, CA. The flag denotes the Thrifty station that was converted to an ARCO station.

Say Something about this...?



Nevo & Whinston: Specifically on the case of Merger Analysis

- Mergers can be very different:
 - Substitutability of products (Post Raisin Bran & Kelloggs Raisin Bran; Thrifty & Shell stations; IAH and Hobby airports)
 - In many mergers, finding a control group problematic. E.g. UA-CO
 - Sometimes “complex superstructure” are actually just more flexible models
 - Einav & Levin to Angrist & Pischke:
 - “Do they seriously think that if the Department of Justice had to review a merger of Microsoft and Yahoo! it should rely on the price effect of past airline or office supply company mergers, or better yet the subset that resulted from chance meetings of chief executive officers or from lunar eclipses?”
- Cannot say anything about welfare effects



Focus on Reduced-Form Research

- Central focus: causal interpretation from articulated research design
 - E.g. in a system of 2 eqns, less focus on say how errors are correlated with each other, and more focus on whether exclusion restrictions are believable (quality of IV / omitted var bias)

Example: Effect of Class Size on Student Achievement

- “Naïve” regressions of student outcomes on class size and student demographics
 - Small class → worse outcomes
 - Possible confounder? Schools group weak students in small classes together.
 - No amount of functional form or error covariance matrix “fixes” are going to fix this problem. It’s a problem in research design.
 - A&P: “Improvements in empirical work have come from many directions...a reduced emphasis on econometric considerations that are not central to a causal interpretation of the main findings”.
- Exogenous variation in size:
 - Tennessee STAR – randomly assigned class sizes for K-3; reduced class size from about 25 to about 15
 - But can one say anything about 45 to 35, or 35 to 25?
 - Israeli class size capped at 40 – regression discontinuity
 - If “naturally” 41 → 2 classes of 20.5; if naturally 39 → size=39
 - But, what if parents respond to large classes by sending kids to another school? What if teachers assigned in high enrollment years are not random? (Keane)

Does Quasi-experimental Research Take the “Econ” out of Econometrics?

- “What’s that have to do with economics?” / “There’s no price theory in there”
 - Does prenatal exposure to radioactive fallout hurt school outcomes?
 - Does drinking reduce academic performance in college?
 - Does having more dormmates of a different race increase the racial integration of social networks?
- “This is just applied statistics.”
 - Best case scenario: sample means
- My View: if question important (e.g. for policy) and economists can answer better than xxxxx-ologists, who cares?
- Profession: heterogeneity in opinions

Credibility Revolution: Does it seek good experiments instead of good questions?

- Do I want to be really, really sure of the answer to a question that doesn't particularly matter?
 - Examples: <not for distribution>
 - Interesting read: *The New Republic* in 2007 “Freaks and Geeks; How Freakonomics is ruining the dismal science”.
- Keep the focus on big questions.
 - Examples: returns to college
- Don't let the causality police make your research uninteresting.

Protagonists – Angrist & Pischke

- “The Credibility Revolution in Empirical Economics: How Better Research Design is Taking the Con out of Econometrics”
- Book: Mostly Harmless Econometrics, 2009
- A&P: “...researchers seek real experiments where feasible, and useful natural experiments if real experiments seem (at least for a time) infeasible.”
“Design-based revolutionaries have notched many successes, putting hard numbers on key parameters of interest to both policymakers and economic theorists. *Imagine what could be learned were a similar wave to sweep the fields of macroeconomics and industrial organization.*”
- Randomization – explicit field experiment
 - Developing countries
 - Move to Opportunity (MTO) – randomly give housing vouchers to low-income people in public housing projects; e.g. outcomes on crime by movers
- Quasi-randomization
 - Brian Jacob – exploit closing of housing projects in Chicago; timing of closing *within* a project; looking at student outcomes

A&P on IO: “Industrial Disorganization”

- Question: Causal effect of a merger? Antitrust officials needs to know when deciding whether to approve.
- A&P contrast Hastings (2004) vs. Nevo (2000)
 - “Can mergers be analyzed using simple, transparent empirical methods that trace a shorter route from facts to findings?”
 - Complain that Nevo uses an “elaborate superstructure”
 - Recognize that regulators must evaluate mergers before they occur, but appear to prefer drawing inferences from previous mergers where one can conduct Hastings-like analyses?
 - Claim that at the very least, structural results should be compared to credible design-based results, where such comparison is available.

“Valued Added” of Structural Models

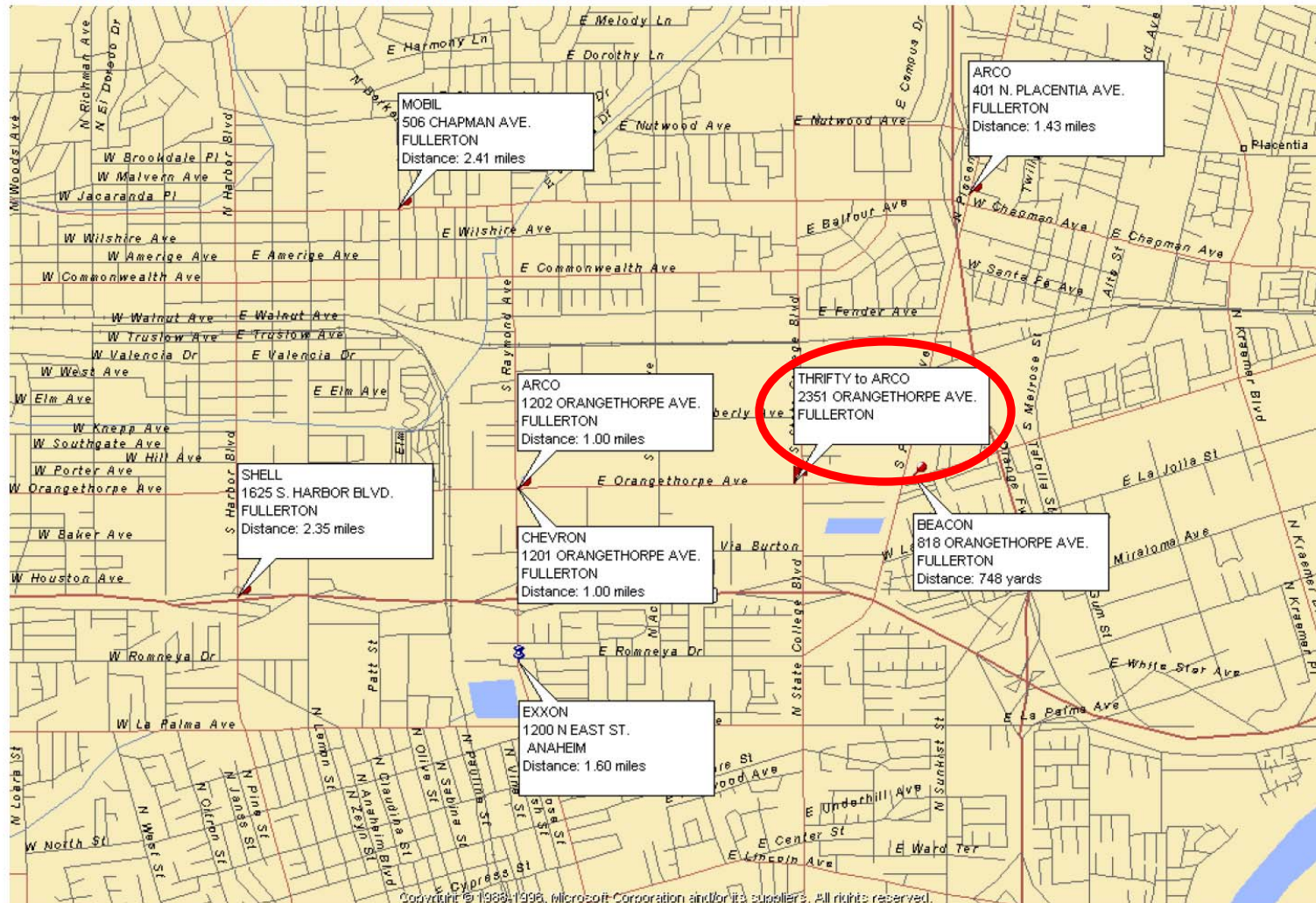
- See Nevo & Whinston for nice discussion
- Structural analysis & credible identification are *complements*, not *substitutes*
- Structural modeling allows us to predict outcomes to changes in environment that are *not-yet-observed* (i.e. don't observe before/after)
 - Lingo: “do counterfactual experiments”
 - Step 1: uses data on past behavior + an economic model to recover fundamental parameters (e.g. preferences, costs)
 - Step 2: use fundamental parameters + model to predict outcomes to environmental changes that have never happened before
 - Example: merger & prices; valuing new products (e.g. minivan)
- Structural model allows us to estimate changes in welfare (using estimated fundamentals such as costs or consumer preferences)
- Call it “numerical theory” if that makes it go down smoother

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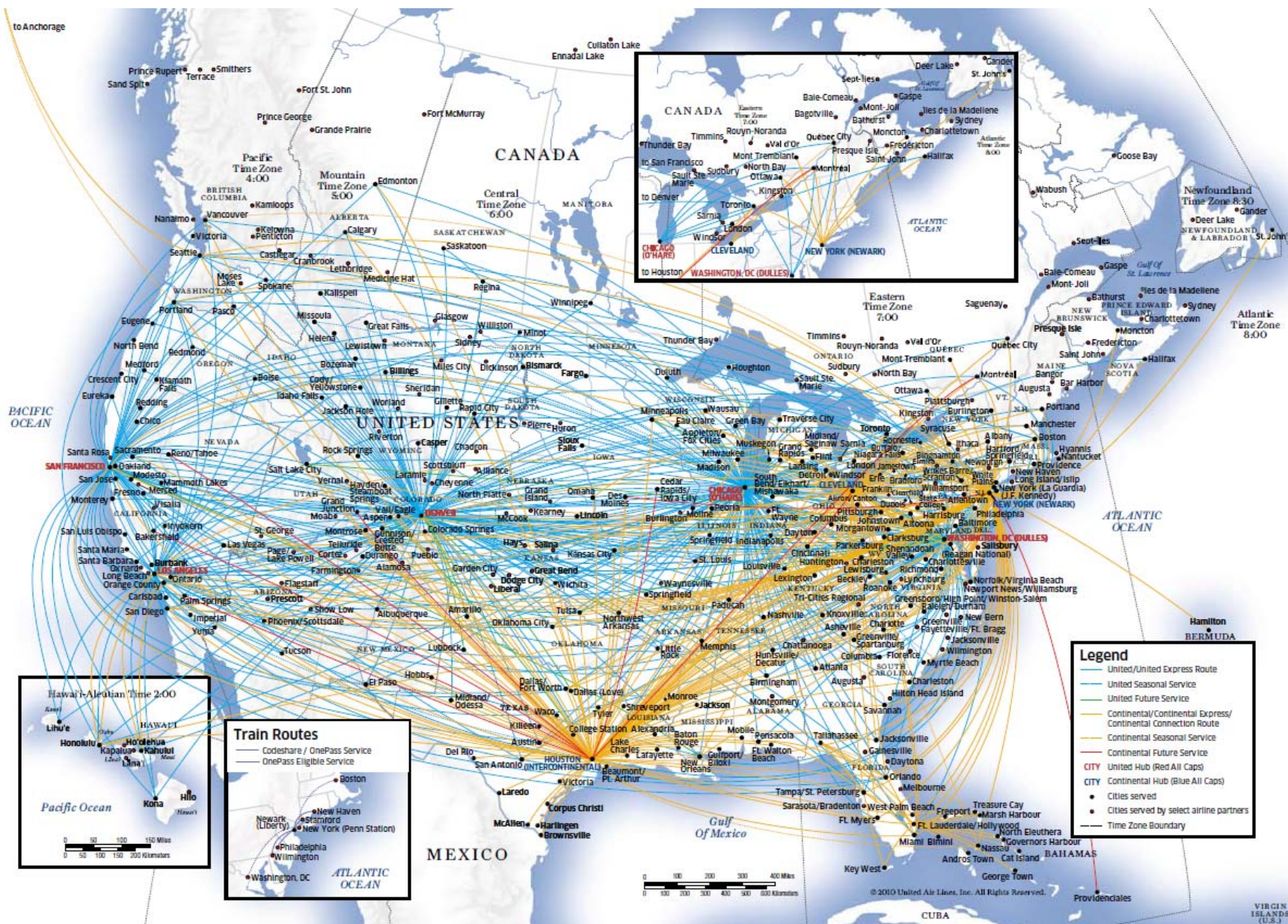
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“It’s not transparent, so I don’t believe it”

- ANY empirical paper (no matter the flavor) makes critical assumptions that may not be testable
 - Author has some idea of robustness to assumptions
 - How to honestly convey to reader?
- Reduced-form models (e.g labor, public) good at this
 - <show example>
- Structural modelers still trying to figure out
 - <see example from Knittel/Metaxoglou>
 - Part of the problem:
 - programming often lots more than running “reg” or “ivreg”
 - Changing each element of vector of starting values?

Showing Robustness in a Reduced-Form Paper

Table 2: The Effect of Exposure to Troubled Children on Reading and Mathematics Test Scores Before and After the Parent Reports Domestic Violence

Specification	1	2	3	4	5	6	7	8	9	10	11	12
Proportion of Peers with Unreported Family Violence	-274.23*** (26.80)	-227.65*** (26.18)	-27.21** (12.21)	-30.21** (12.40)	-28.85** (12.27)	-37.11*** (9.96)	-35.01*** (9.17)	-27.56*** (9.33)	-24.48** (10.02)	-22.19* (12.22)	-23.70** (11.44)	-23.53* (12.65)
Proportion of Peers with Reported Family Violence	-163.16*** (25.59)	-192.34*** (23.85)	-1.60 (11.53)	-3.62 (12.14)	-3.45 (12.14)	3.56 (9.76)	5.36 (9.78)	-2.19 (10.26)	-4.34 (11.12)	-4.11 (10.86)	1.15 (8.72)	-4.43 (9.90)
Own Unreported Family Violence	-14.15*** (1.28)	-13.58*** (1.27)	-11.09*** (1.29)	-11.13*** (1.29)	-11.11*** (1.28)	-4.48*** (1.08)	-4.47*** (1.08)	-4.39*** (1.08)	-4.35*** (1.08)	-5.81** (2.80)	0.04 (1.76)	-0.32 (0.57)
Own Reported Family Violence	-11.45*** (1.24)	-11.90*** (1.22)	-9.33*** (1.27)	-9.36*** (1.28)	-9.35*** (1.28)	-3.36*** (0.95)	-3.37*** (0.95)	-3.44*** (0.94)	-3.40*** (0.94)	-1.60 (2.33)		-0.9 (0.52)
Observations	42,478	42,478	42,478	42,478	42,478	42,478	42,478	42,478	42,478	26,922	42,478	21,148
Year Fixed Effects	No	Yes	Yes	Yes	-	-	-	-	-	-	-	-
School Fixed Effects	No	No	Yes	-	-	-	-	-	-	-	-	-
School-Grade Fixed Effects	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grade-Year Fixed Effects	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Controls	No	No	No	No	No	Yes	Yes	Yes	Yes	-	-	Yes
Cohort Controls	No	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
School-Grade-specific linear time trends	No	No	No	No	No	No	No	Yes	No	No	No	No
School-Year Fixed Effects	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Sibling Fixed Effects	No	No	No	No	No	No	No	No	No	Yes	-	No
Individual Fixed Effects	No	No	No	No	No	No	No	No	No	No	Yes	No
Lagged Test Score	No	No	No	No	No	No	No	No	No	No	No	Yes

Notes: Each column reports results from a separate regression. Individual controls include own reported and unreported domestic violence, gender, race, median family income, and subsidized lunch status. Cohort controls include average gender, race, subsidized lunch, and size of cohort by school-by-grade-by-year. Standard errors in parentheses are two-way clustered at the school by cohort and school by year level in Specifications 1 through 8; for computational reasons standard errors in specifications including sibling or individual fixed effects are clustered at the school by cohort level.

* Significant at the 10% level

** Significant at the 5% level

*** Significant at the 1% level

“Robustness” in Classic BLP paper

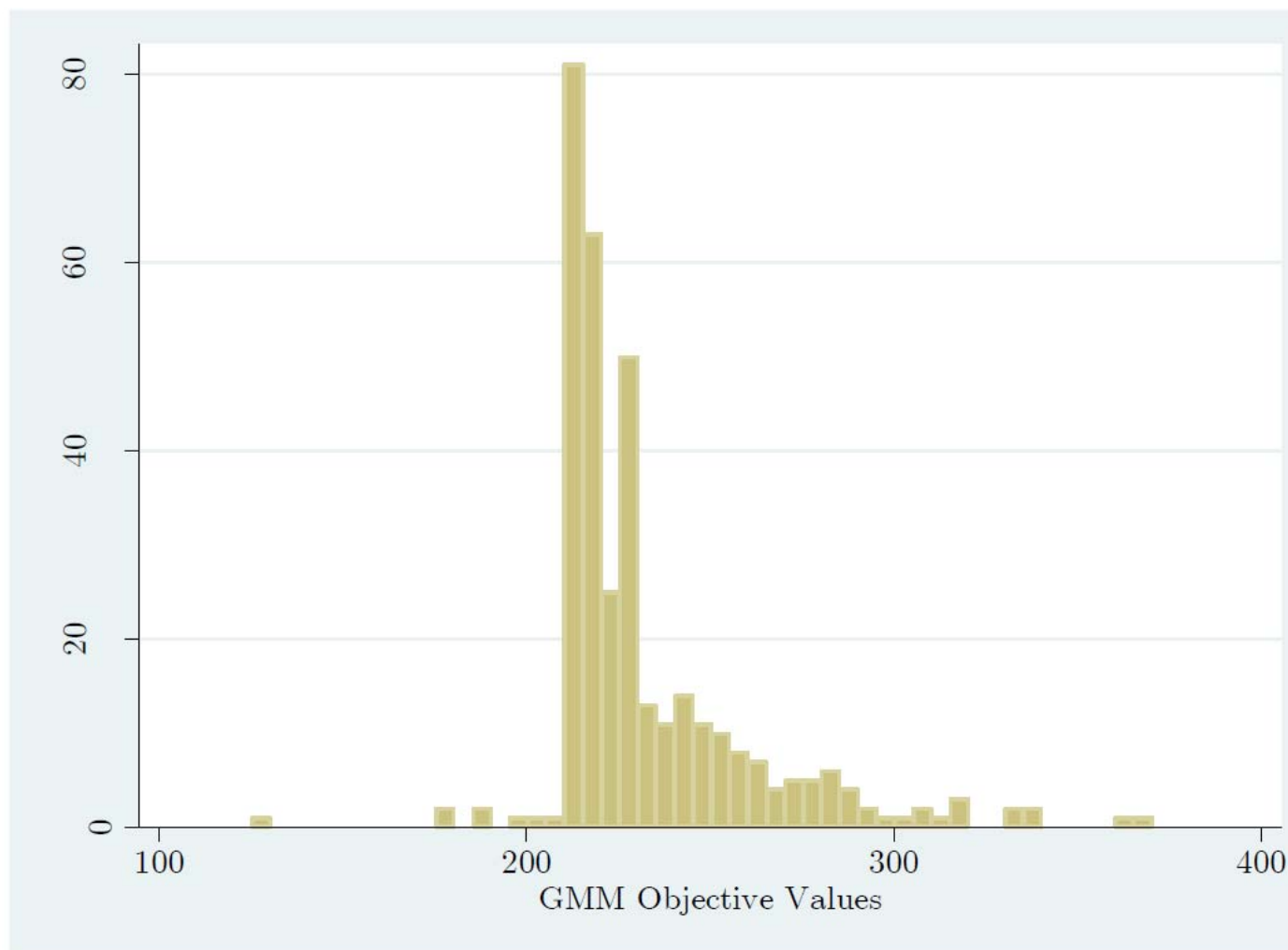


Figure 2: A histogram of GMM objective values for converged algorithms using the automobile data. This truncates the upper 10% of the converged GMM objective values.

Tradeoff Between “Credibility” and “Consequence”

- Intense focus on credible identification can lead to researchers to “only” answer questions where there is suitable exogeneity
 - A&P response: studies of specific settings are useful if they speak to more general behavior
 - Einav & Levin: may not want to always follow markets with good data or established methods
- Sims: “If applied economists narrow the focus of their research and critical reading to various forms of pseudo-experimental analysis, the profession loses a good part of its ability to provide advice about the effects and uncertainties surrounding policy issues.”
 - Extreme example: global climate change
 - A&P seem to view accumulation of incremental steps as answer: will that get you there for certain “big” issues?

Einav & Levin Response

- “Empirical Industrial Organization: A Progress Report”
 - Read it!
- A&P would have you estimate cereal price elasticities by looking for periods when Froot Loops pricing algorithm malfunctioned (exogenously)
- Do we think that economic theory is a useful lens through which to view markets?
 - Pricing – hopefully a qualified yes
 - Education – an equilibrium model of student learning incorporating peer effects and optimal school, student, teacher, and parent decisions? Maybe “statistical approach” better
- Best to use both compelling identifying variation and economic models
 - Echoes the Nevo-Winston point that it’s not “either-or”

Looking Forward...

- Both approaches will continue but each will incorporate insights of other
 - E.g. structural papers with a slide saying “Identification Strategy”. Pay close attention to what generates the variation in the data.
- “Elaborate superstructure” made more transparent
 - E.g. Nevo’s “Practitioner’s Guide”
- Researchers with more “quasi-experimental” leanings will get more comfortable with (some) structural models so they can do counterfactual experiments
- Differences will persist across fields
 - IO – mix of structural and reduced-form; “silly” structural will be filtered out
 - Public/labor – still a strong inclination to reduced-form, but some movement
 - e.g. Hastings on school choice, retirement investments; Handel – health insur.

My Personal Strategy

- My personal preference: I prefer papers that have some reasonable chance to land in a top journal. Full stop.
- Follow good questions and use whatever econometrics tools are needed to get a credible “answer”
 - I use different approaches depending on the data I have
 - E.g. social networks at TAMU (no “quasi-randomization”) vs. Rice (“quasi-randomization”)
 - You will need a toolkit that has different empirical strategies, so you can apply the appropriate tool when you encounter different situations
- “consumption preferences” vs. ideology
 - ➔ you need to learn your preferences
- If a researcher is truly interested in answering questions, then by the time an applied academic is “middle-aged”, (s)he should have papers that are a mix of structural and reduced-form. Empirically not the case.
If not,
 - a good reason is: everyone has a comparative advantage
 - a bad reason is: they think their methodology is the “right” one.